Surbhit Wagle (He/Him)

Computational Neuroscientist

Date and Place of birth: Novermber, 21, 1994, Khategaon, India
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WORK EXPERIENCE

PhD Researcher

University Hospital

September 2023 – Ongoing Bonn, Germany

University Medical Center

October 2021 – August 23 Mainz, Germany

Max Planck Institute for Brain Research

July 2020 – September 2021 Frankfurt, Germany

- Successfully developed a **statistical model** of mRNA and protein trafficking kinetics in neurons using **differential equations**
- Successfully deployed deep learning and algorithms based software in **Python** for **processing, analysis and visualization bioimaging data** which led to significant reduction in time and efforts for experimental research.
- Effectively employed **parameter optimization** to reduce the complexity of extensive, **high-dimensional data** and **data-driven modeling**.
- Effectively implemented **numerical simulations** in **Python** for integrating **partial differential equations**.
- Successfully organized the **German Neuroscience Olympiad** for two years and **designed** the problem set for the Olympiad.
- Led a student-run initiative to encourage **collaboration, team-building, and budget management** between students with diverse backgrounds and expertise.

Junior Research Assistant

National Center for Biological Sciences/Institute for Stem Cell Science and Regenerative Medicine

November 2018 – May 2020 Bangalore, India

- Effectively optimized a **large-scale, complex ($\sim$ 500 reactions and $\sim$ 3000 parameters), data-driven, biochemical reaction network** models using **Hierarchical optimization strategy**.
- Effectively deployed a web-application to store large, curated dataset on biochemical signaling experiments in **PHP** and **Python**.
- Successfully deployed a **REST framework pipeline** in **Python** for **parallel** and **batch optimization** jobs on the Neuroscience Gateway's (NSG) **HPC** supercomputers at the University of California, San Diego, USA.
- Efficaciously managed a **cross-institute, collaborative consortium**, **SANKET**, for **neuroscience research in health and disease** in India.

Research Assistant

Indian Institute of Technology

July 2016 – October 2018 Kanpur, India

- Successfully developed an suite with **five video games for android** utilizing **Unity3D game engine** and deployed the app on **Google Play**.
- Effectively employed **iterative development strategy** to improve initial version of games with active user testing.
- Efficaciously designed **large-scale database in SQL** and implemented **back-end** for storing user data and displaying it using Model View Controller design pattern.
- Effectively employed **iterative development methodologies** like **Scrum** in a team of UI/UX designers, software developers, psychologists, and medical professionals

EDUCATION

PhD. in Computational Neuroscience

University of Bonn

Expected December 2024

Master's degree in Biological Sciences and BioEngineering, GPA: 1.8/4

Indian Institute of Technology Kanpur

Sept 2016 – Oct 2018

Bioinformatics | Cell signaling

Bachelor's degree in Computer Engineering, GPA: 2.3/4

Institute of Engineering and Technology

Sept 2012 – June 2016

Algorithms and Data structures | Software Development

TECHNICAL SKILLS

Python ●●●●●
(TensorFlow,PyTorch, pandas, Scikit-Learn)

Deep/Machine Learning ●●●●●
(CNN,RNN, Autoencoders, regression)

Modeling and Simulations ●●●●●
(Finite Difference Methods, Numerical Integration)

Software Development ●●●●○
(Python, C/C++, C#, Unity)

Data Visualization ●●●●●
(Matplotlib, Seaborn)

Version control ●●●●○
(Git,Bitbucket)

Mathematics ●●●○○
(Differential Equations, Statistics)

Web-development ●●●●○
(Django, PHP, Drupal, HTML, CSS, JavaScript)

Databases management ●●●●○
(MySQL, SQLite)

Image analysis ●●●●○
(Scikit-Image, Pytorch)

Operating systems ●●●●○
(Windows, Unix/Linux, Mac OS)

MANAGEMENT SKILLS

Analytical thinking | Problem-Solving | Quality control | Interdisciplinary mindset | Conflict Management | Effective Communication | Collaboration and Teamwork | Presentation

RELEVANT WORKSHOPS/COURSES

- *CNEURO: Theoretical and Computational Neuroscience Summer School*. Tsinghua University, China, Online, 2020.
- *First Italian Summer School on Geometric Deep Learning (GeoDL2022)*. Università di Chieti-Pescara, Italy, 2022.
- *Deep Learning for Image Analysis*. European Molecular Biology Laboratory, Heidelberg, Germany, 2023.
- *Bio-inspired Deep Learning*. Joachim Herz Stiftung, Guntersblum, Germany, 2024.
- *Making Mathematical Models of Human Disease Reproducible Hackathon*. BioMed X Institute, Heidelberg, Germany, 2024.

PUBLICATIONS

Journal Articles

- **Surbhit Wagle** et al. "Development and testing of a game-based digital intervention for working memory training in autism spectrum disorder". In: *Sci. Rep.* 11.1 (July 2021), p. 13800. DOI: 10.1038/s41598-021-93258-w.
- Arka Ghosh et al. "An Unguided, Computerized Cognitive Behavioral Therapy Intervention (TreadWill) in a Lower Middle-Income Country: Pragmatic Randomized Controlled Trial". In: *J Med Internet Res* 25 (Apr. 2023), e41005. ISSN: 1438-8871. DOI: 10.2196/41005.
- **Surbhit Wagle** et al. "Computational insights into mRNA and protein dynamics underlying synaptic plasticity rules". In: *Molecular and Cellular Neuroscience* 125 (2023), p. 103846. ISSN: 1044-7431. DOI: https://doi.org/10.1016/j.mcn.2023.103846.
- Maximilian F Eggl et al. "SpyDen: Automating molecular and structural analysis across spines and dendrites". In: *bioRxiv* (2024), pp. 2024–06.

Conference presentations

- *70th Annual National Conference*, Indian Psychiatry Society, Ranchi, India, 2018, (Poster).
- *Bernstein Conference*, Bernstein Network, Berlin, Geramny, 2022, (Attended).
- *Bernstein Conference, Bernstein Network*. Berlin, Geramny, 2023, (Poster).
- *Emergent AI workshop*. Emergent Algorithmic Intelligence, Carl Zeiss Stiftung, Nierstein, Germany, 2023, (Invited talk).
- *FENS Forum*, Federation of European Neuroscience Societies, Paris, France, 2022, (Poster).
- *FENS Forum*, Federation of European Neuroscience Societies, Online, 2020, (Attended).
- *International Symposium*. Collaborative Research Center 1080 Molecular and Cellular Mechanisms of Neural Homeostasis, Frankfurt am Main, Germany, 2024, (Poster).
- *No Garland Neuroscience Conference*. , Indian Institute for Science Education and Research Pune, India, 2020, (Poster).
- *SfN Conference*. ,Society for Neuroscience, Washington DC, USA, 2023, (Poster).
- *SfN Conference*. , Society for Neuroscience, San Diego, CA, USA, 2022, (Poster).
- *SfN Conference*. , Society for Neuroscience, Chicago, IL, USA, 2022, (Poster).

LANGUAGES

English ●●●●●

Hindi ●●●●●

Marathi ●●●●●

German (A1) ●○○○○

AWARDS

Cooperation Project Grant

Collaborative Research Centre 1080, Frankfurt am Main, Germany

October 2022 - March 2023

Awarded with 10,000 € grant for a period of 6 months to work on a collaborative, interdisciplinary project

Starting Scholarship

Goethe Research Academy for Early Career Researchers, Frankfurt am Main, Germany

July 2021 - September 2021

Awarded with 3000 € grant over 3 months to work remotely during corona pandemic

Graduate Aptitude Test for Engineering Fellowship

Indian Institute of Science, Bangalore, India

July 2016 - June 2018

Awarded to top *1%* scholars for pursuing higher education in engineering and sciences in India